

Masaki Imagawa

Curriculum Vitae

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Contact Information

Institute of Mathematics, Academia Sinica
No. 1, Sec. 4, Roosevelt Road, Da-an, Taipei 106319, Taiwan
Email: imagawa@as.edu.tw Website: m-imagawa.github.io

Employment

Postdoctoral Fellow Institute of Mathematics, Academia Sinica, Taipei, Taiwan	Aug. 2025–Present
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Education

Ph.D. in Informatics , Kyoto University Dissertation: Convergence analysis of elliptic regularization for advection equations in a bounded domain Advisor: Hiroshi Fujiwara	Apr. 2022–Jul. 2025
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Master of Informatics , Kyoto University	Apr. 2020–Mar. 2022
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B.Sc. in Mathematics , Kyoto University	Apr. 2016–Mar. 2020
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Publications

Peer-reviewed Articles

Masaki Imagawa and Daisuke Kawagoe, A revisit on well-posedness of a boundary value problem of a stationary advection equation without the separation condition, *J. Math. Anal. Appl.* **155** (2025), no. 2, part 2, Art. 129695.
[doi:10.1016/j.jmaa.2025.129695](https://doi.org/10.1016/j.jmaa.2025.129695)

Conference Proceedings

Masaki Imagawa and Yuusuke Iso, Sharp estimates in singular perturbation applied to the first order partial differential equations in a bounded domain, in *2025 Days on Diffraction*, IEEE, 2025, 90–95.
[doi:10.1109/DD66835.2025.11263497](https://doi.org/10.1109/DD66835.2025.11263497)

Preprints

Masaki Imagawa and Daisuke Kawagoe,
On strong convergence of an elliptic regularization with the Neumann boundary condition applied to a stationary advection equation, [arXiv:2303.17904](https://arxiv.org/abs/2303.17904), 2023.

Presentations

Talks

Stability and Boundary Interaction of Viscous Shock Waves for the Burgers Equation 19th EASIAM Annual Meeting, Jeju, Korea	Aug. 2026
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Stability of Shock Waves for the Viscous Burgers Equation with Boundary Interactions Taipei Postdoc Seminar, Taipei	Apr. 2026
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Stability of Viscous Shock Waves for the Burgers Equation in a Half-Line Computational Mechanics Seminar, Kyoto	Apr. 2026
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Asymptotic stability of traveling waves for scalar viscous conservation laws with boundary Applied Mathematics Freshman Seminar 2025, Kyoto	Nov. 2025
Convergence estimates for singular perturbation of advection equations with mixed boundary conditions Kyoto University Applied Mathematics Seminar, Kyoto	Jul. 2025
Convergence analysis of perturbed advection equations in a bounded domain Numerical Analysis Seminar, The University of Tokyo, Tokyo	Jul. 2025
Sharp estimates in singular perturbation applied to the first order partial differential equations in a bounded domain Days on Diffraction, online	Jun. 2025
Boundary value problems of advection equations and its regularized problems Analysis Seminar, Academia Sinica, online	Mar. 2025
On a Boundary Value Problem of a First-Order PDE and Its Regularization Problems 2024 Japan-Taiwan Joint Workshop on Numerical Analysis and Inverse Problems, Kyoto	Mar. 2024
Convergence Rates of Elliptic/Parabolic Regularization Applied to Advection Equations with Boundary Conditions The 5th Workshop on Recent Development of Mathematical Fluid Dynamics and Hyperbolic Conservation Laws, Jeju	Sep. 2023
On the Optimal Convergence Rate of Elliptic Regularization Methods Applied to Advection Equations The 19th Joint Meeting of JSIAM Activity Groups, Okayama Joint work with Daisuke Kawagoe and Yuusuke Iso	Mar. 2023
On Convergence Rates of an Elliptic Regularization Applied to a Stationary Advection Equation The 13th Taiwan-Japan Joint Workshop for Young Scholars in Applied Mathematics, online	Mar. 2023
On Convergence Estimation of Elliptic Regularization Applied to Boundary Value Problem of Advection Equations 2022 Japan-Taiwan Joint Online Workshop on Numerical Analysis and Inverse Problem, online	Mar. 2022
Numerical Computation of Advection Equations Using Elliptic Regularization Annual Meeting of the Japan Society for Industrial and Applied Mathematics, online Joint work with Daisuke Kawagoe, Hiroshi Fujiwara and Yuusuke Iso	Sep. 2021
Poster	
Convergence rates of elliptic regularization applied to advection equations with boundary conditions NCTS–Kyoto Mathematics Symposium, Taipei	Dec. 2023
On the Convergence Rate of Singularly Perturbed Solutions to the Steady Advection Equation The 17th ICT Innovation, Kyoto University, Kyoto Joint work with Daisuke Kawagoe	Feb. 2023